19. Conductivity of a saturated solution of a sparingly soluble salt AB (1:1 electrolyte) at 298K is 1.85×10^{-5} S m^{-1}. Solubility product of the salt AB at 298K $\left(\Lambda^{\circ}_{m}\right)_{AB} = 14\times10^{-3}$ S m^2 mol^{-1}.

a) 5.7×10^{-12} b) 1.32×10^{-12} c) 7.5×10^{-12} d) 1.74×10^{-12}

20. In the electrochemical cell: $Zn|ZnSO_4\ (0.01M)\|CuSO_4(1.0M)|Cu$, the emf of this Daniel cell is E_1. When the concentration of $ZnSO_4$ is changed to 1.0M and that $CuSO_4$ changed to 0.01M, the emf changes to E_2. From the above, which one is the relationship between E_1 and E_2?

a) $E_1 < E_2$ b) $E_1 > E_2$ c) $E_2 \geq E_1$ d) $E_1 = E_2$

21. Consider the change in oxidation state of Bromine corresponding to different emf values as shown in the diagram below:

$$BrO_4^- \xrightarrow{1.82V} BrO_3^- \xrightarrow{1.5V} HBrO \xrightarrow{1.595V} Br_2 \xrightarrow{1.0652V} Br^-$$

Then the species undergoing disproportionation is

a) Br_2 b) BrO_4^- c) BrO_3^- d) HBrO

22. For the cell reaction

$2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$

$E^{\circ}_{cell} =$ 0.24V at 298K. The standard Gibbs energy (Δ, G°) of the cell reactions is :

a) -46.32 KJ mol^{-1} b) -23.16 KJ mol^{-1} c) 46.32 KJ mol^{-1} d) 23.16 KJ mol^{-1}

23. A certain current liberated 0.504gm of hydrogen in 2 hours. How many grams of copper can be liberated by the same current flowing for the same time through copper sulphate solution

a) 31.75 b) 15.8 c) 7.5 d) 63.5

24. A gas X at 1 atm is bubbled through a solution containing a mixture of $1MY^-$ and $1MZ^-$ at 25°C. If the reduction potential of Z>Y>X, then

a) Y will oxidize X and not Z b) Y will oxidize Z and not X

d) Y will oxidize both X and Z d) Y will reduce both X and Z

25. Cell equation : $A + 2B^- \rightarrow A^{2+} + 2B$;

$A^{2+} + 2e^- \rightarrow A$ $E^{\circ} = +0.34V$ and $\log_{10}K = 15.6$ at 300K for cell reactions find E° for $B^+ + e^- \rightarrow B$ (AIIMS – 2018)

a) 0.80 b) 1.26 c) -0.54 d) -10.94

Short Answer Questions

1. Define anode and cathode
2. Why does conductivity of a solution decrease on dilution of the solution
3. State Kohlrausch Law. How is it useful to determine the molar conductivity of weak electrolyte at infinite dilution.

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